MIL-M-0038510/25C(USAF)
1 December 1975
USED IN LIEU OF
MIL-M-38510/25B
18 June 1975

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL,
LOW POWER, COUNTERS, MONOLITHIC SILICON

This limited coordination military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/25B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, low power, binary and decade counters for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

| M38510 | /025 | 01 | S | T | D |
|---|---|---|---|---|---|
| Military designator | Detail specification | Device type (1.2.1) | Device class (1.2.2) | Case outline (1.2.3) | Lead finish (3.3) |

1.2.1 Device type. The device type shall be as shown in the following:

| Device type | Circuit | Similar commercial type |
|---|---|---|
| 01 | Decade counter | 54L90 |
| 02 | 4-bit binary counter | 54L93A |

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range - - - - - - - - - - - - | 0 Vdc to 8.0 Vdc |
| Input voltage range- - - - - - - - - - - - - | 0 Vdc to 5.5 Vdc |
| Storage temperature range- - - - - - - - - | -65°C to 150°C |
| Lead temperature (soldering, 10 seconds) - | 300°C |

1.4 Recommended operating conditions.

| | |
|---|---|
| Ambient operating temperature range- - - - | -55°C to 125°C |
| Supply voltage - - - - - - - - - - - - - - - | 4.5 Vdc minimum to 5.5 Vdc maximum |
| Minimum high level input voltage - - - - - - | 2.0 Vdc |
| Maximum low level input voltage- - - - - - - | 0.7 Vdc |

FSC 5962

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight, and physical dimensions. The design, construction, weight, and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics shall be as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C, and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)



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TABLE I. Electrical characteristics tests (device type 01).

| Test No. 1/ 2/ | Characteristics | Symbol | Conditions | Limits 3/ | | Delta (Δ) 1/ | Units |
|---|---|---|---|---|---|---|---|
| | | | | Min | Max | | |
| 1 | Input voltage required to ensure a high-level input (each input) | $V_{IH}$ | $V_{CC}$ = 4.5 V | 2 | | | V |
| 2 | Input voltage required to ensure a low-level input (each input) | $V_{IL}$ | $V_{CC}$ = 4.5 V | | 0.7 | | V |
| 3* | High-level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V, $V_{IH}$ = 2 V<br>$V_{IL}$ = 0.7 V, $I_{OH}$ = -100 µA | 2.4 | | ±.12 | V |
| 4* | Low-level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V, $V_{IH}$ = 2 V<br>$V_{IL}$ = 0.7 V, $I_{OL}$ = 2 mA | | 0.3 | ±.03 | V |
| 5 | Low-level input current at any reset input | $I_{IL1}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 0.3 V | -0.06 | -.18 | | mA |
| 6 | Low-level input current at A input | $I_{IL2}$ | | -0.12 | -.54 | | mA |
| 7 | Low-level input current at BD input | $I_{IL3}$ | | -0.36 | -1.08 | | mA |
| 8* | High-level input current at any reset input | $I_{IH1}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 2.4 V | | 10 | ±1.0 | µA |
| 9* | High-level input current at A input | $I_{IH2}$ | | | 30 | ±3.0 | µA |
| 10* | High-level input current at BD input | $I_{IH3}$ | | | 60 | ±6.0 | µA |
| 11 | High-level input current at any reset input | $I_{IH4}$ | $V_{CC}$ = 5.5 V<br>$V_{IN}$ = 5.5 V | | 100 | | µA |
| 12 | High-level input current at A input | $I_{IH5}$ | | | 300 | | µA |
| 13 | High-level input current at BD input | $I_{IH6}$ | | | 600 | | µA |
| 14 | Short-circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V | -3 | -15 | | mA |
| 15 | Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V | | 7.2 | | mA |

1/ Delta computations shall be performed on parameters marked with an asterisk (*) with data read and recorded.

2/ Test sequence is optional with the manufacturer.

3/ Unless otherwise specified, the limits apply at $T_A$ = -55°C to +125°C.





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TABLE I. Electrical characteristics tests (device type 02). - Continued

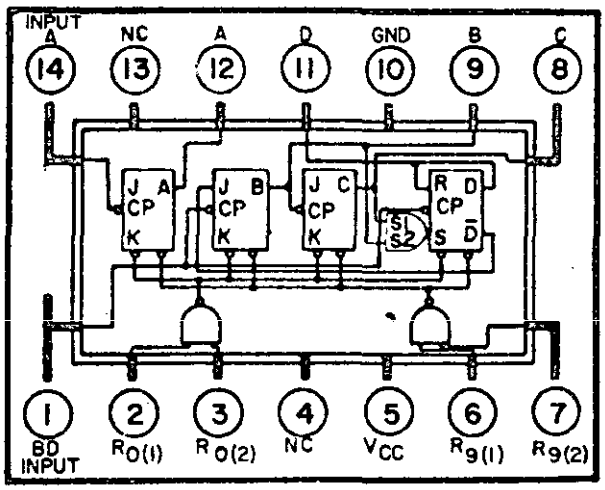
| Test No. 1/ 2/ | Characteristics | Symbol | Conditions | Limits 3/ | | Delta (Δ) 1/ | Units |
|---|---|---|---|---|---|---|---|
| | | | | Min | Max | | |
| 1 | Input voltage required to ensure a high-level input | $V_{IH}$ | $V_{CC}$ = 4.5 V | 2 | | | V |
| 2 | Input voltage required to ensure a low-level input | $V_{IL}$ | $V_{CC}$ = 4.5 V | | 0.7 | | V |
| 3* | High-level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V, $I_{OH}$ = -100 µA<br>$V_{IL}$ = 0.7 V, $V_{IH}$ = 2 V | 2.4 | | ±.12 | V |
| 4* | Low-level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V, $I_{OL}$ = 2 mA<br>$V_{IL}$ = 0.7 V, $V_{IH}$ = 2 V | | 0.3 | ±.03 | V |
| 5* | High-level input current at any reset input | $I_{IH1}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V | | 10 | 10X or 1 µA | µA |
| | | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V | | 100 | | µA |
| 6* | High-level input current at A or B input | $I_{IH3}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V | | 20 | 10X or 2 µA | µA |
| | | $I_{IH4}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V | | 200 | | µA |
| 7* | Low-level input current at any reset input | $I_{IL1}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.3 V | -0.06 | -0.18 | ±.01 | mA |
| 8* | Low-level input current at A or B input | $I_{IL2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.3 V | -0.12 | -0.36 | ±.05 | mA |
| 9 | Short-circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V, $V_{OUT}$ = 0 V | -3 | -15 | | mA |
| 10 | Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V | | 6.6 | | mA |

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at $T_A$ = -55°C to +125°C.



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DEVICE TYPE 01




DEVICE TYPE 02

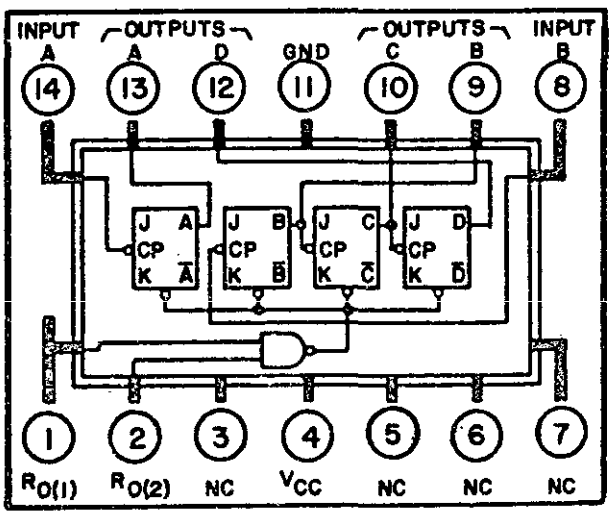



FIGURE 1. Logic diagram and terminal connections.



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BCD COUNT SEQUENCE
(see note 1)

| Count | Output | | | |
|---|---|---|---|---|
| | D | C | B | A |
| 0 | L | L | L | L |
| 1 | L | L | L | H |
| 2 | L | L | H | L |
| 3 | L | L | H | H |
| 4 | L | H | L | L |
| 5 | L | H | L | H |
| 6 | L | H | H | L |
| 7 | L | H | H | H |
| 8 | H | L | L | L |
| 9 | H | L | L | H |

Bi-Quinary (5-2)
(see note 2)

| Count | Output | | | |
|---|---|---|---|---|
| | A | D | C | B |
| 0 | L | L | L | L |
| 1 | L | L | L | H |
| 2 | L | L | H | L |
| 3 | L | L | H | H |
| 4 | L | H | L | L |
| 5 | H | L | L | L |
| 6 | H | L | L | H |
| 7 | H | L | H | L |
| 8 | H | L | H | H |
| 9 | H | H | L | L |

RESET/COUNT FUNCTION TABLE

| RESET INPUTS | | | | OUTPUT | | | |
|---|---|---|---|---|---|---|---|
| $R_{o(1)}$ | $R_{o(2)}$ | $R_{9(1)}$ | $R_{9(2)}$ | D | C | B | A |
| H | H | L | X | L | L | L | L |
| H | H | X | L | L | L | L | L |
| X | X | H | H | H | L | L | H |
| X | L | X | L | COUNT | | | |
| L | X | L | X | COUNT | | | |
| L | X | X | L | COUNT | | | |
| X | L | L | X | COUNT | | | |

DEVICE TYPE 01

NOTES:
1. Output A is connected to input BD for BCD count.
2. Output D is connected to input A for BiQuinary count.
3. H = high level, L = low level, X = irrelevant.

FIGURE 2. Truth tables.



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| Count | Output | | | |
|---|---|---|---|---|
| | D | C | B | A |
| 0 | L | L | L | L |
| 1 | L | L | L | H |
| 2 | L | L | H | L |
| 3 | L | L | H | H |
| 4 | L | H | L | L |
| 5 | L | H | L | H |
| 6 | L | H | H | L |
| 7 | L | H | H | H |
| 8 | H | L | L | L |
| 9 | H | L | L | H |
| 10 | H | L | H | L |
| 11 | H | L | H | H |
| 12 | H | H | L | L |
| 13 | H | H | L | H |
| 14 | H | H | H | L |
| 15 | H | H | H | H |

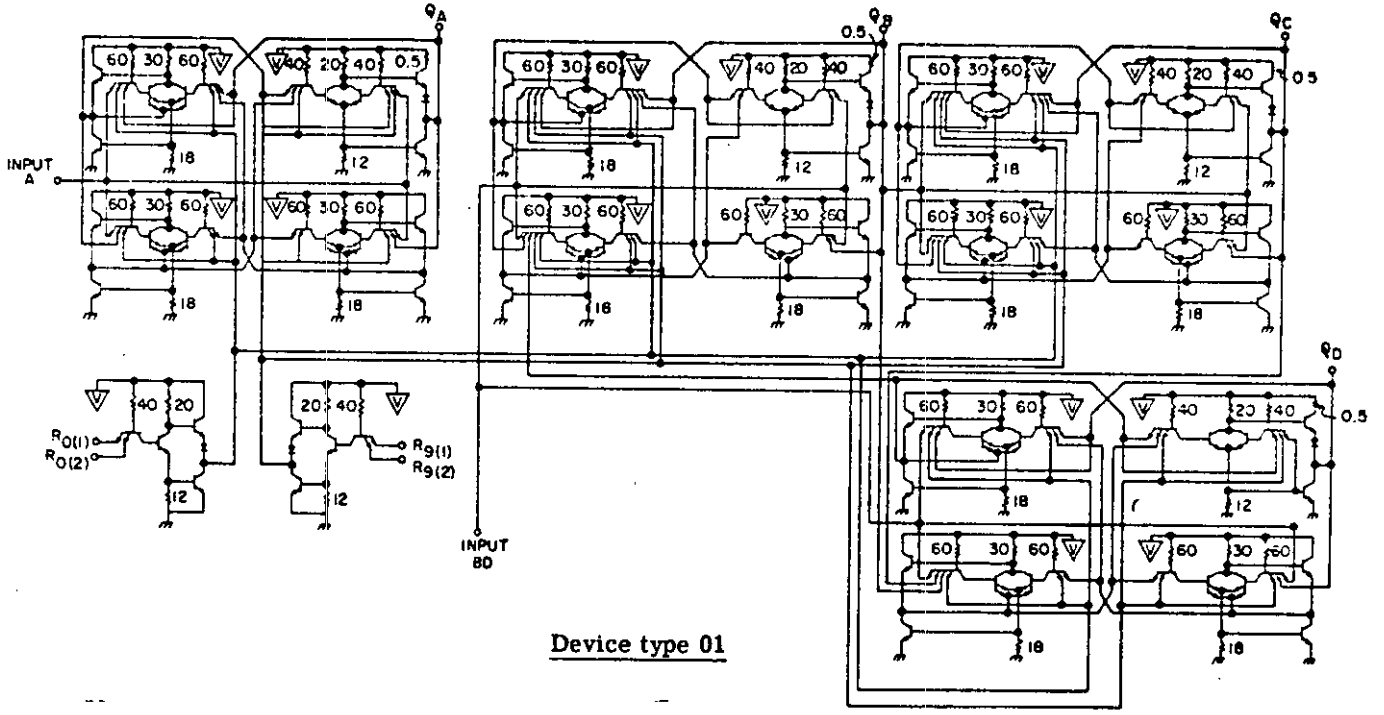
DEVICE TYPE 02

NOTES:
1. Output A connected to input B.
2. To reset all outputs to logical L, both $R_{0(1)}$ and $R_{0(2)}$ inputs must be at logical H.
3. Either (or both) reset inputs $R_{0(1)}$ and $R_{0(2)}$ must be at logical L to count.

FIGURE 2. Truth tables - (continued).



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Device type 01

NOTE: Resistor values shown are nominal in kilohms.
▽: $V_{CC}$ bus

FIGURE 3. Schematic circuits.



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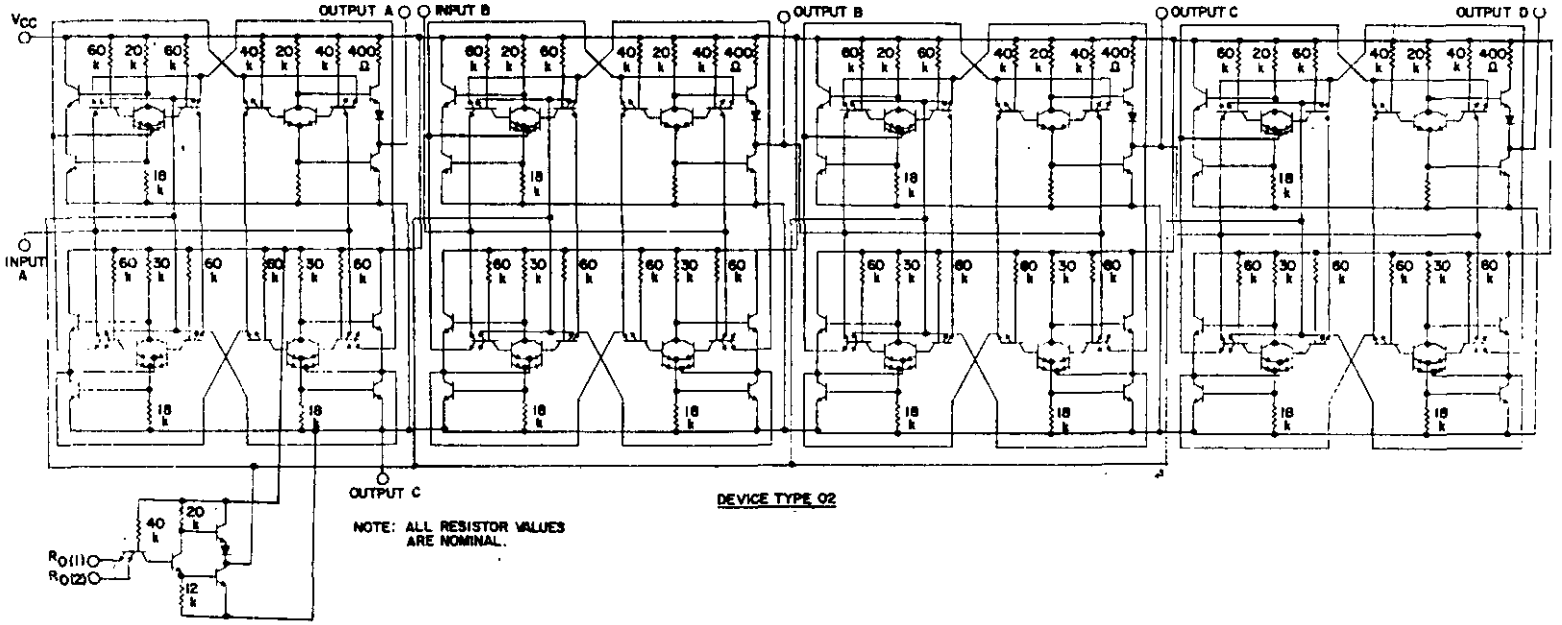
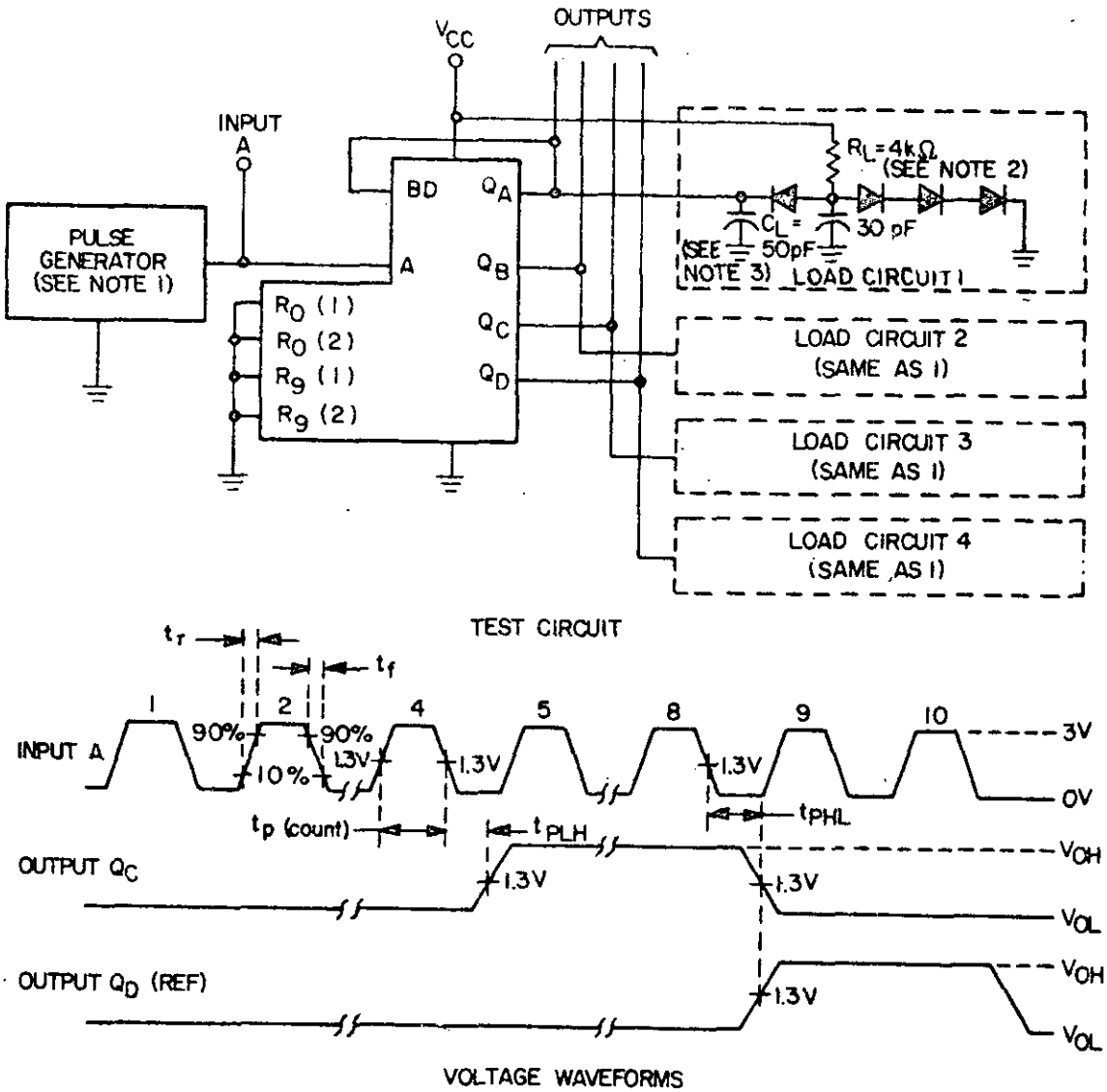



FIGURE 3. Schematic circuits - Continued.





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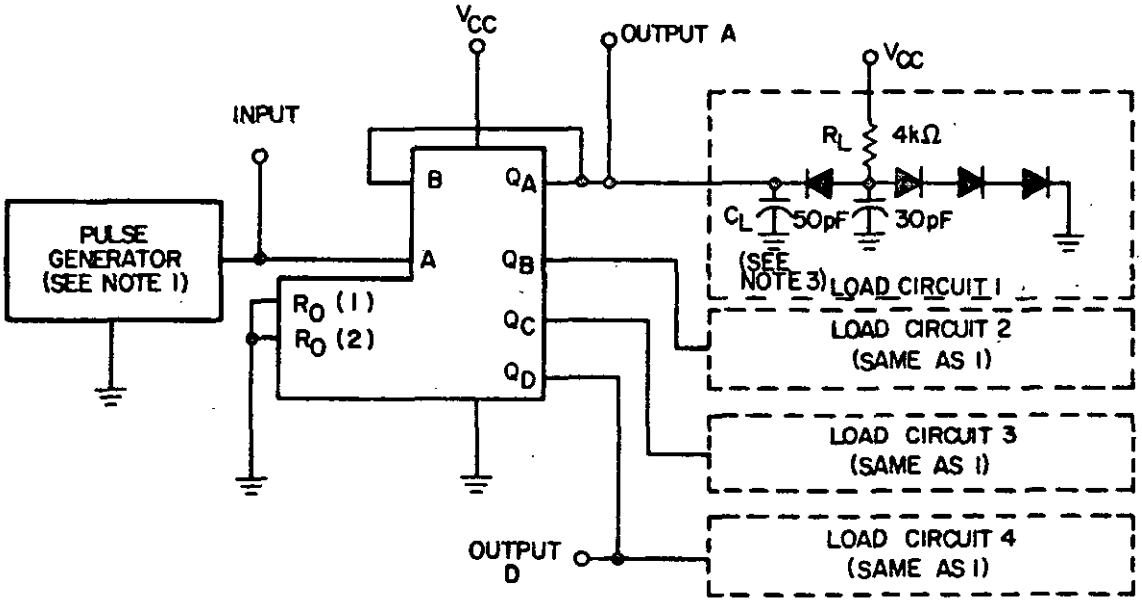



NOTES:
1. The pulse generator has the following characteristics: $t_p \geq 200$ ns, PRR = 500 kHz, $Z_{out} \approx 50\Omega$, $t_r \leq t_f \leq 15$ ns.
2. All diodes are 1N3064 or equivalent.
3. $C_L$ includes probe and jig capacitance.
4. When testing $f_{MAX}$: PRR = 3 MHz for subgroup A-9 and 2.5 MHz for subgroups A-10 and A-11. Omit load circuits 2, 3 and 4 and remove connection from output $Q_A$ to input BD.

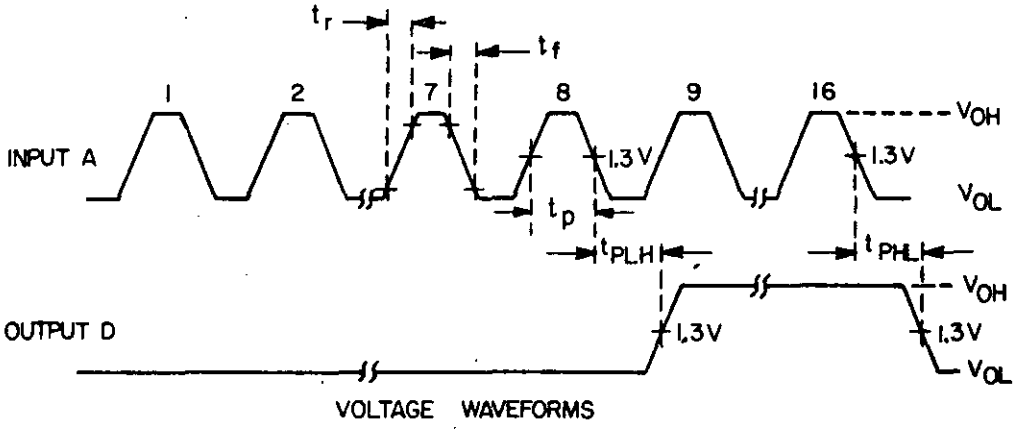
FIGURE 4. Switching time test circuits and waveforms for device type 01.



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TEST CIRCUIT




VOLTAGE WAVEFORMS

NOTES:
1. Pulse generator has the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, PRR = 500 kHz, $Z_{out} \approx 50\Omega$.
2. Voltage values are with respect to network ground terminal.
3. $C_L$ = 50 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
4. All diodes are 1N3064 or equivalent.
5. When testing $f_{MAX}$; PRR = 3 MHz for tests at 25°C, and 2.5 MHz for tests at 125°C and -55°C. Also, omit load circuits 2, 3, and 4 and remove connection from output A to input B.

FIGURE 5. Switching time test circuits and waveforms for device type 02.



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3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figures 4 and 5 shall be performed at 25°C. All devices shall meet the limits specified in table II.

TABLE II. Switching characteristics tests (device type 01).

| Characteristics 1/ | Symbol | Conditions | Test figure | Temp. | Limits | | Units |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Frequency of input count pulses | $f_{MAX}$ | $V_{CC}$ = 5.0 V<br>$C_L$ = 50 pF<br>$R_L$ = 4 kΩ<br>±5% | 4 | 25°C | 3.0 | | MHz |
| | | | | -55°C, 125°C | 2.5 | | |
| Propagation delay time high-to-low level | $t_{PHL}$ | | | 25°C | | 340 | ns |
| | | | | -55°C, 125°C | | 510 | |
| Propagation delay time low-to-high level | $t_{PLH}$ | | | 25°C | | 340 | ns |
| | | | | -55°C, 125°C | | 510 | |

1/ Input A to output QA for $f_{MAX}$ tests.
Input A to output QD for $t_{PHL}$ and $t_{PLH}$ tests.

TABLE II. Switching characteristics tests (device type 02).

| Characteristics | Symbol | Conditions | Test figure | Temp. | Limits | | Units |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Frequency of input clock pulses | $f_{MAX}$ | $V_{CC}$ = 5.0 V<br>$C_L$ = 50 pF<br>$R_L$ = 4 kΩ<br>±5% | 5 | 25°C | 3.0 | | MHz |
| | | | | 125°C, -55°C | 2.5 | | |
| Propagation delay, low-to-high level, input count pulse to output | $t_{PLH}$ | | | 25°C | | 450 | ns |
| | | | | 125°C, -55°C | | 675 | |
| Propagation delay, high-to-low level, input count pulse to output | $t_{PHL}$ | | | 25°C | | 450 | ns |
| | | | | 125°C, -55°C | | 675 | |

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. Quality assurance shall be in accordance with MIL-M-0038510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-0038510.



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6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-55)



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